

JAVA PROGRAMMING LAB (BCS-351)

B.Tech CSE 3rd SEMESTER

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Assignment 1: Hello World, Variables and Arithmetic

Aim: Print text, store values in variables and calculate with arithmetic operators.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 1 - Question 1
 */
class Main {
    public static void main(String[] args) {
        System.out.println("Hello World");
        System.out.println("Welcome to Java");
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment1\Question1> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment1\Question1> java Main
Hello World
Welcome to Java
PS D:\Java-Lab-Assignments\Lovish\Assignment1\Question1>
```

Assignment 1: Hello World, Variables and Arithmetic

Aim: Print text, store values in variables and calculate with arithmetic operators.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 1 - Question 2
 */
class Main {
    public static void main(String[] args) {
        int roll = 101;
        double percentage = 84.5;
        char section = 'B';
        String name = "Asha";

        System.out.println("Roll: " + roll);
        System.out.println("Percentage: " + percentage);
        System.out.println("Section: " + section);
        System.out.println("Name: " + name);
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment1\Question2> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment1\Question2> java Main
Roll: 101
Percentage: 84.5
Section: B
Name: Asha
PS D:\Java-Lab-Assignments\Lovish\Assignment1\Question2>
```

Assignment 1: Hello World, Variables and Arithmetic

Aim: Print text, store values in variables and calculate with arithmetic operators.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 1 - Question 3
 */
class Main {
    public static void main(String[] args) {
        int a = 17;
        int b = 5;

        System.out.println("Sum: " + (a + b));
        System.out.println("Difference: " + (a - b));
        System.out.println("Product: " + (a * b));
        System.out.println("Quotient: " + (a / b));
        System.out.println("Remainder: " + (a % b));
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment1\Question3> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment1\Question3> java Main
Sum: 22
Difference: 12
Product: 85
Quotient: 3
Remainder: 2
PS D:\Java-Lab-Assignments\Lovish\Assignment1\Question3>
```

Assignment 1: Hello World, Variables and Arithmetic

Aim: Print text, store values in variables and calculate with arithmetic operators.

Question 4: Question4

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 1 - Question 4
 */
class Main {
    public static void main(String[] args) {
        final double PI = 3.14;
        int radius = 7;

        double area = PI * radius * radius;
        double perimeter = 2 * PI * radius;

        System.out.println("Area: " + area);
        System.out.println("Perimeter: " + perimeter);
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

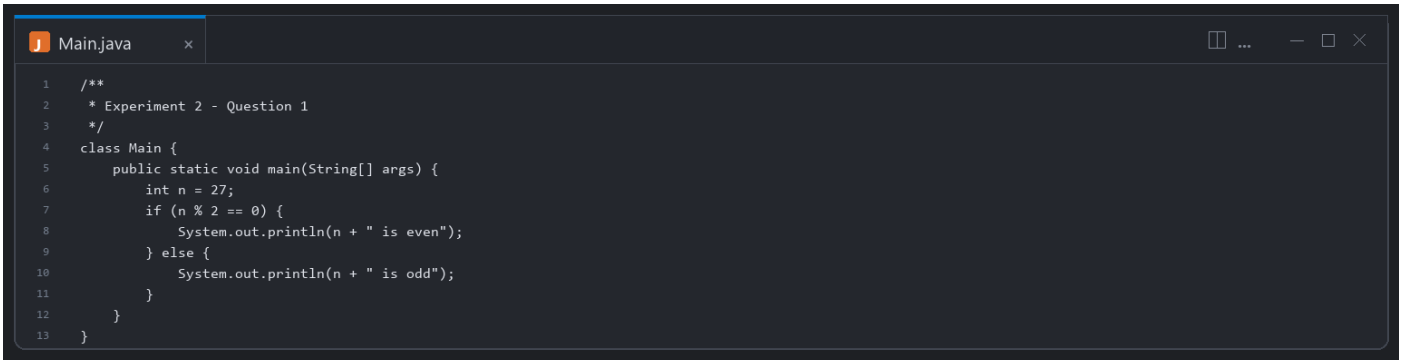
```
PS D:\Java-Lab-Assignments\Lovish\Assignment1\Question4> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment1\Question4> java Main
Area: 153.86
Perimeter: 43.96
PS D:\Java-Lab-Assignments\Lovish\Assignment1\Question4>
```

Assignment 2: Decisions and Loops with Numbers

Aim: Use if, else if and loops to make decisions and repeat steps with whole numbers.

Question 1: Question1

Code screenshot (IDE-style source view)



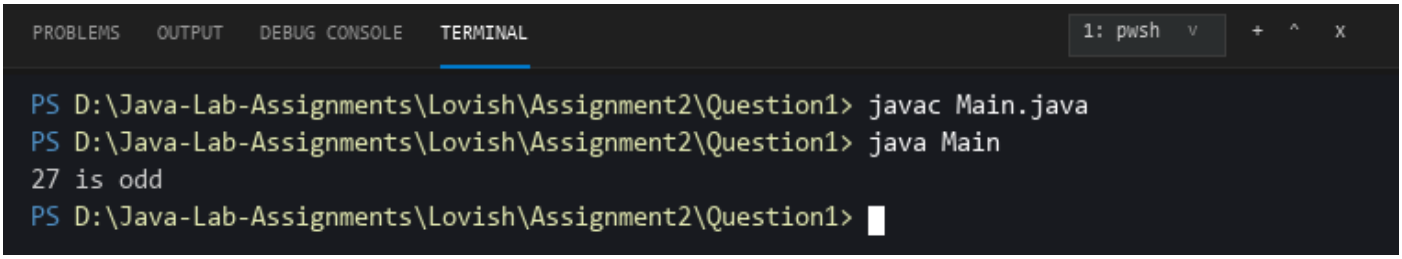
Main.java

```
/**
 * Experiment 2 - Question 1
 */
class Main {
    public static void main(String[] args) {
        int n = 27;
        if (n % 2 == 0) {
            System.out.println(n + " is even");
        } else {
            System.out.println(n + " is odd");
        }
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot



PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment2\Question1> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment2\Question1> java Main
27 is odd
PS D:\Java-Lab-Assignments\Lovish\Assignment2\Question1>
```

Assignment 2: Decisions and Loops with Numbers

Aim: Use if, else if and loops to make decisions and repeat steps with whole numbers.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 2 - Question 2
 */
class Main {
    public static void main(String[] args) {
        int marks = 83;
        if (marks >= 90) {
            System.out.println("Grade: A");
        } else if (marks >= 75) {
            System.out.println("Grade: B");
        } else {
            System.out.println("Grade: C");
        }
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment2\Question2> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment2\Question2> java Main
Grade: B
PS D:\Java-Lab-Assignments\Lovish\Assignment2\Question2>
```

Assignment 2: Decisions and Loops with Numbers

Aim: Use if, else if and loops to make decisions and repeat steps with whole numbers.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 2 - Question 3
 */
class Main {
    public static void main(String[] args) {
        int year = 2024;
        if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
            System.out.println("Leap year");
        } else {
            System.out.println("Not a leap year");
        }
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL                1: pwsh  v  +  ^  x

PS D:\Java-Lab-Assignments\Lovish\Assignment2\Question3> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment2\Question3> java Main
Leap year
PS D:\Java-Lab-Assignments\Lovish\Assignment2\Question3>
```

Assignment 2: Decisions and Loops with Numbers

Aim: Use if, else if and loops to make decisions and repeat steps with whole numbers.

Question 4: Question4

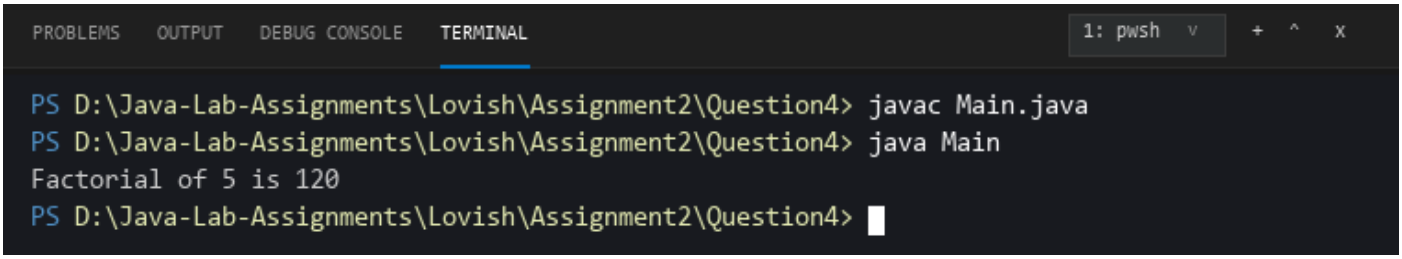
Code screenshot (IDE-style source view)



Input

No runtime input required; the values are initialized in the program.

Output screenshot

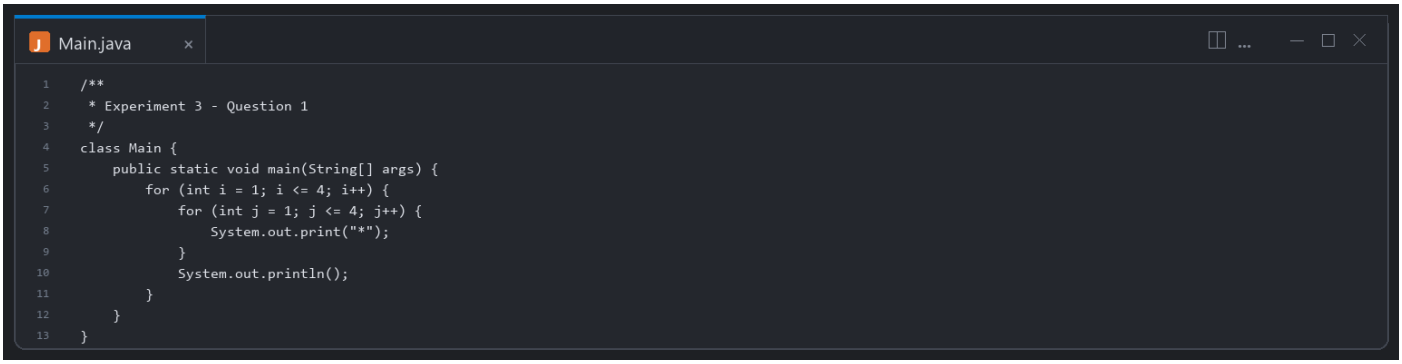


Assignment 3: Star and Number Patterns

Aim: Use a loop inside another loop to print stars and numbers in rows.

Question 1: Question1

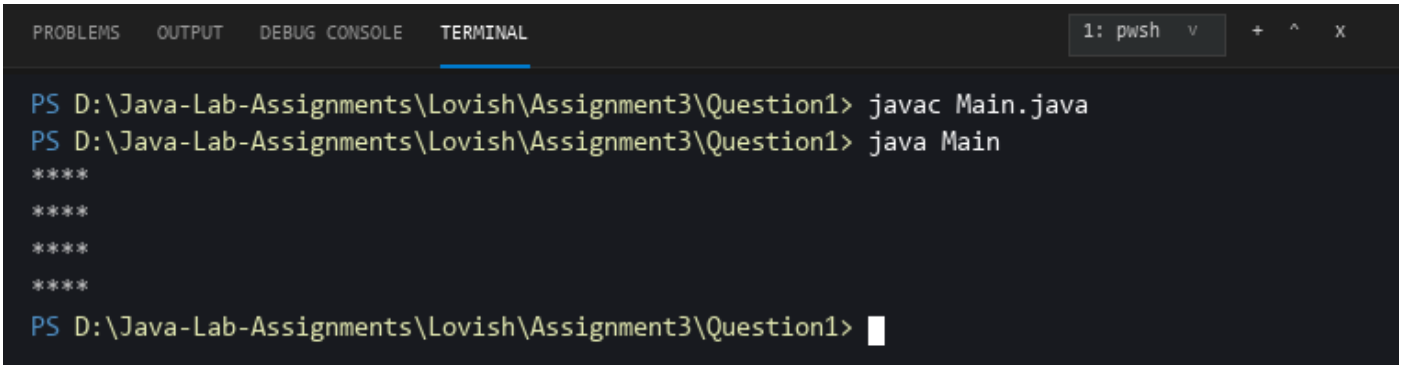
Code screenshot (IDE-style source view)



Input

No runtime input required; the values are initialized in the program.

Output screenshot

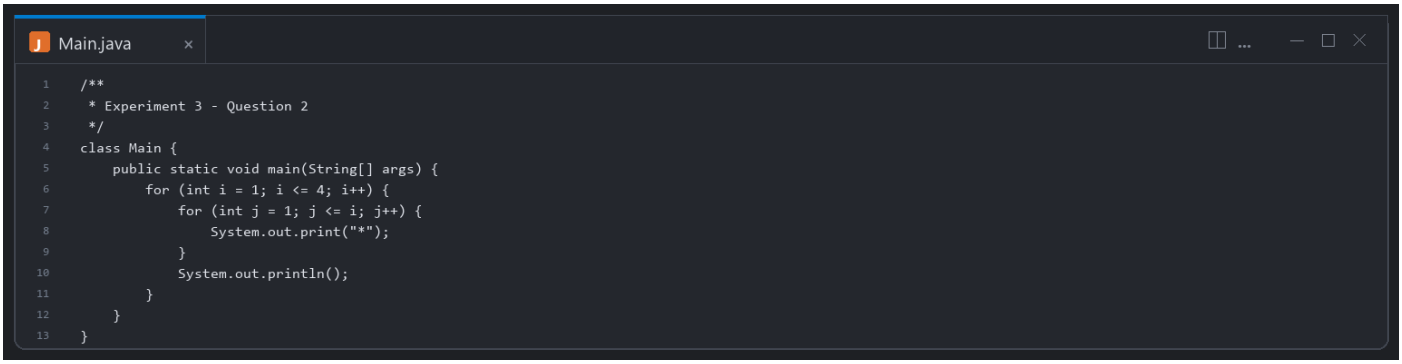


Assignment 3: Star and Number Patterns

Aim: Use a loop inside another loop to print stars and numbers in rows.

Question 2: Question2

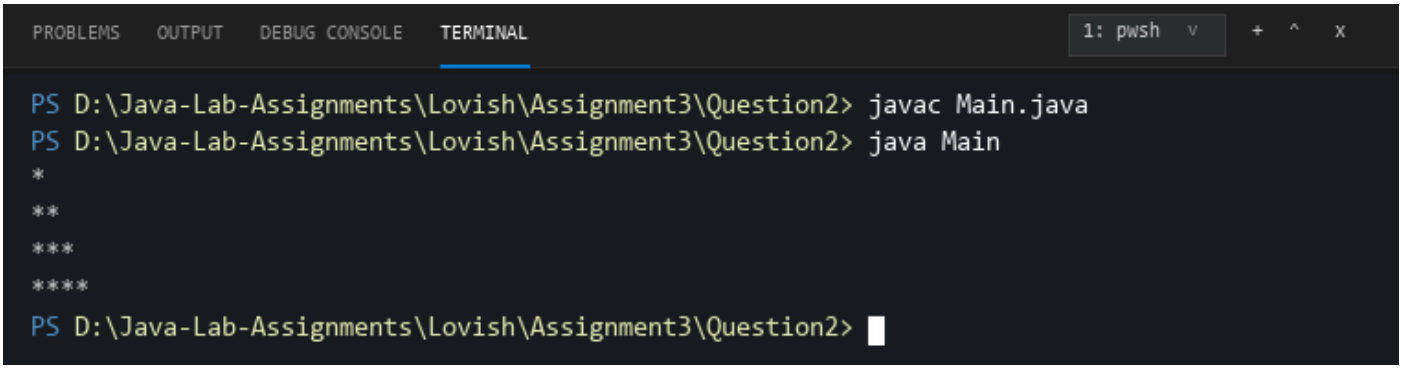
Code screenshot (IDE-style source view)



Input

No runtime input required; the values are initialized in the program.

Output screenshot

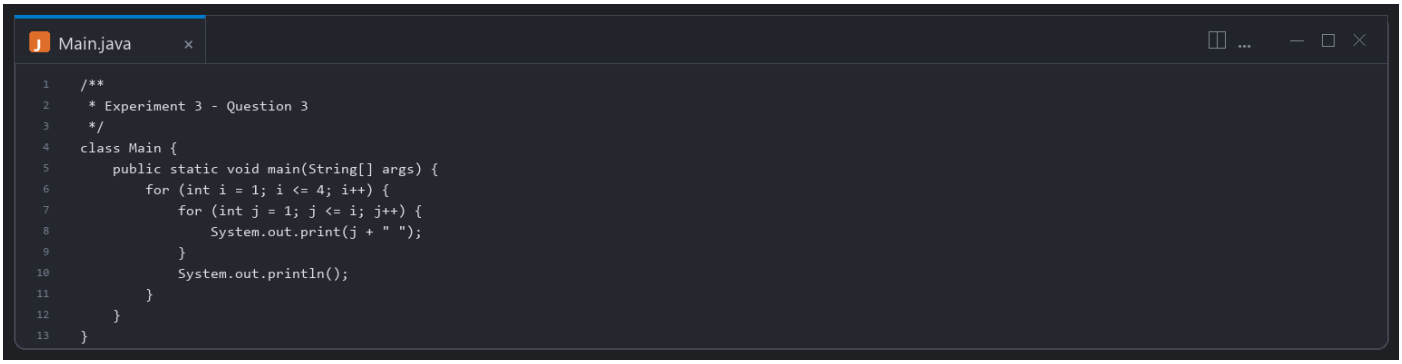


Assignment 3: Star and Number Patterns

Aim: Use a loop inside another loop to print stars and numbers in rows.

Question 3: Question3

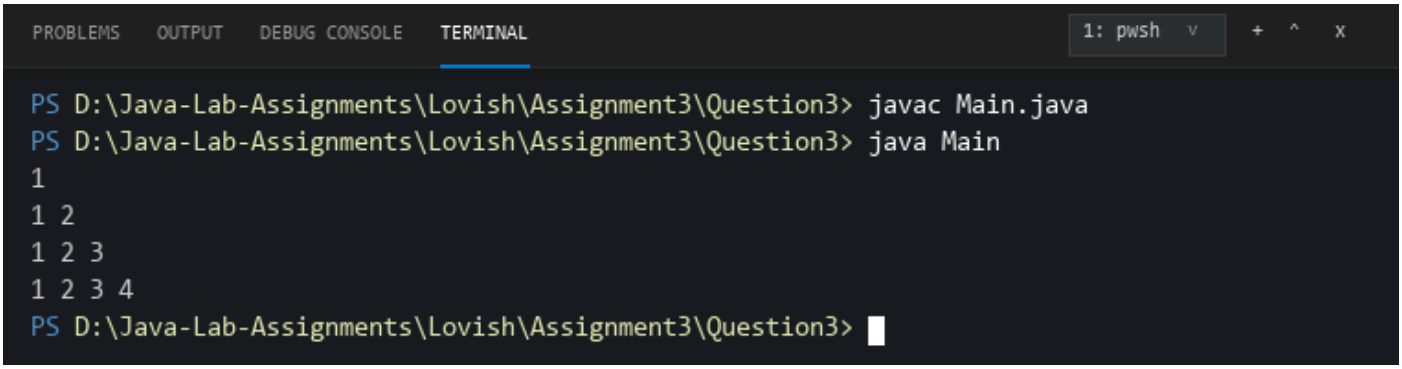
Code screenshot (IDE-style source view)



Input

No runtime input required; the values are initialized in the program.

Output screenshot

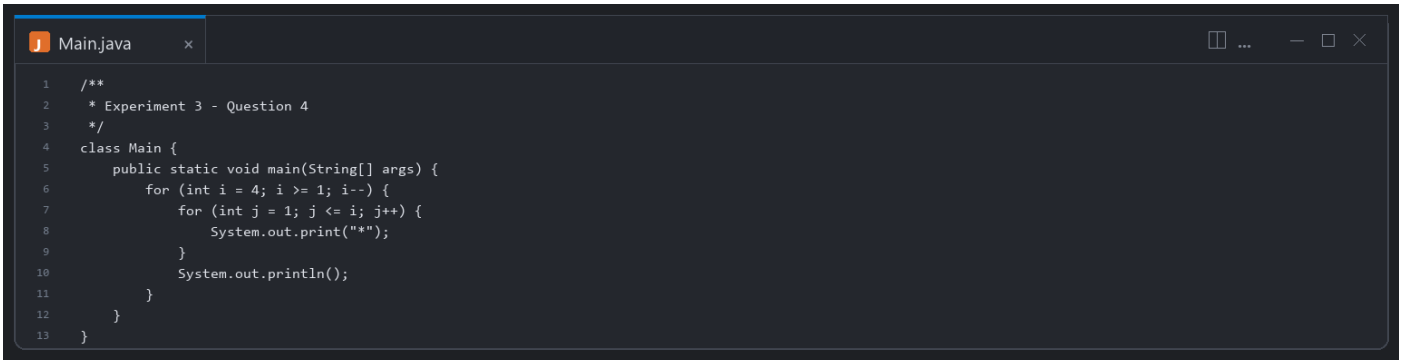


Assignment 3: Star and Number Patterns

Aim: Use a loop inside another loop to print stars and numbers in rows.

Question 4: Question4

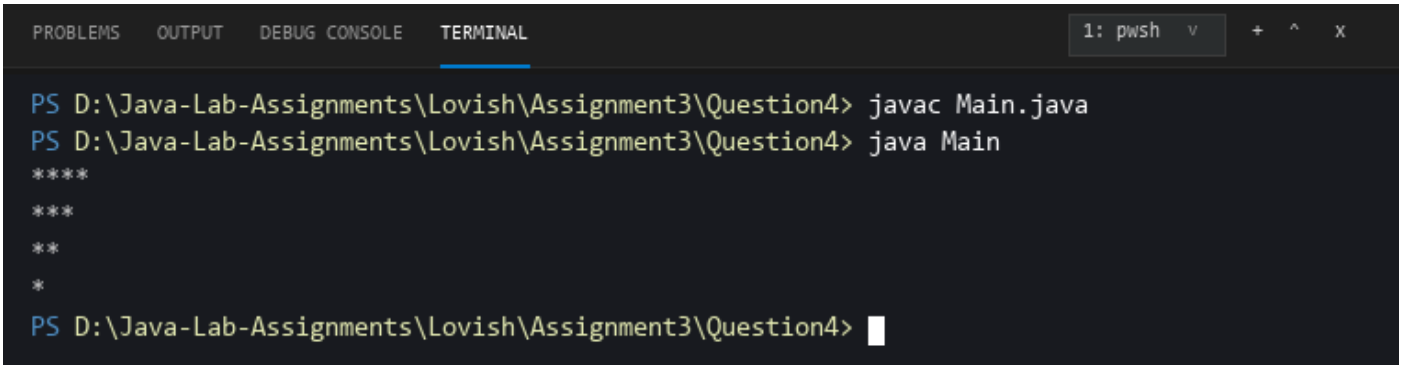
Code screenshot (IDE-style source view)



Input

No runtime input required; the values are initialized in the program.

Output screenshot

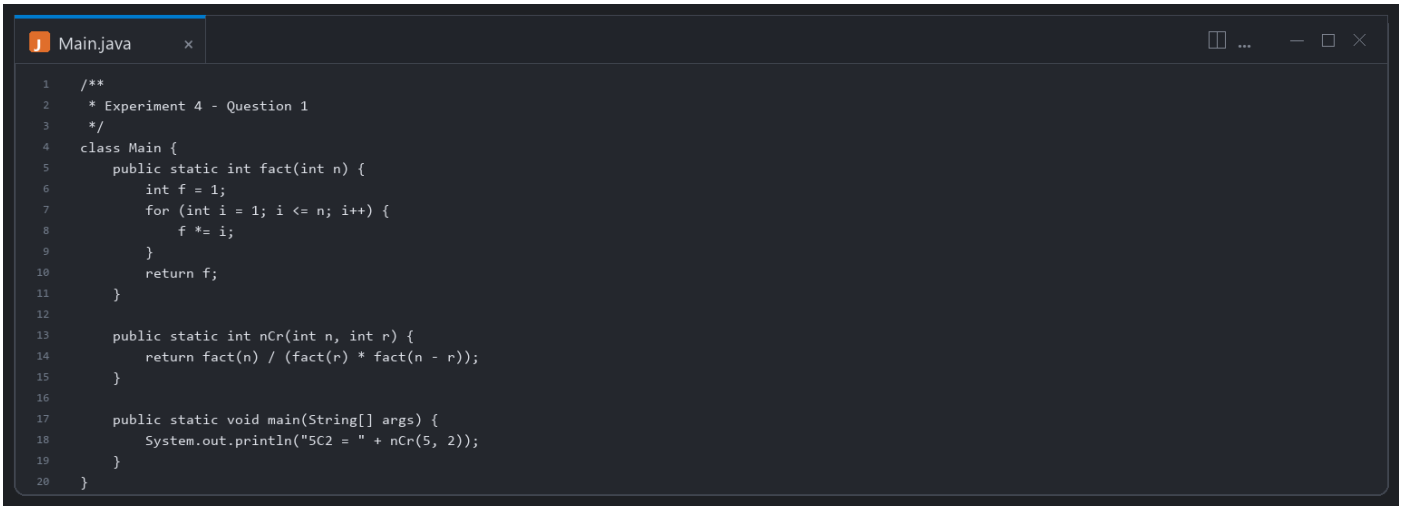


Assignment 4: Writing Functions

Aim: Write functions with parameters and return values, and call them from main.

Question 1: Question1

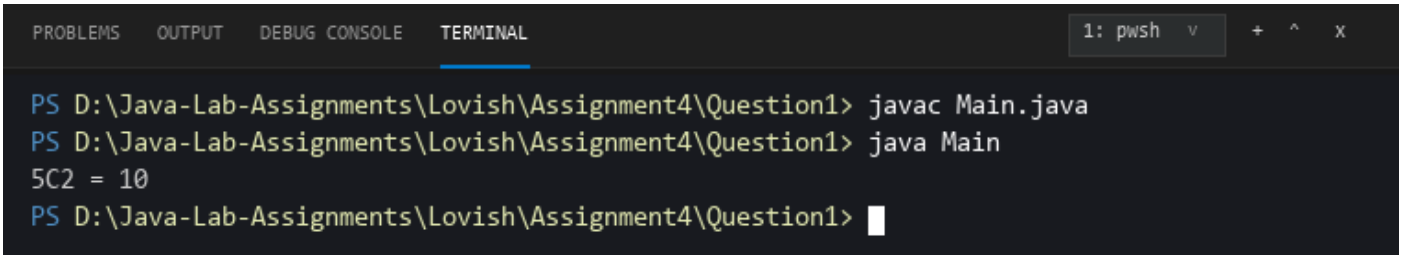
Code screenshot (IDE-style source view)



Input

No runtime input required; the values are initialized in the program.

Output screenshot

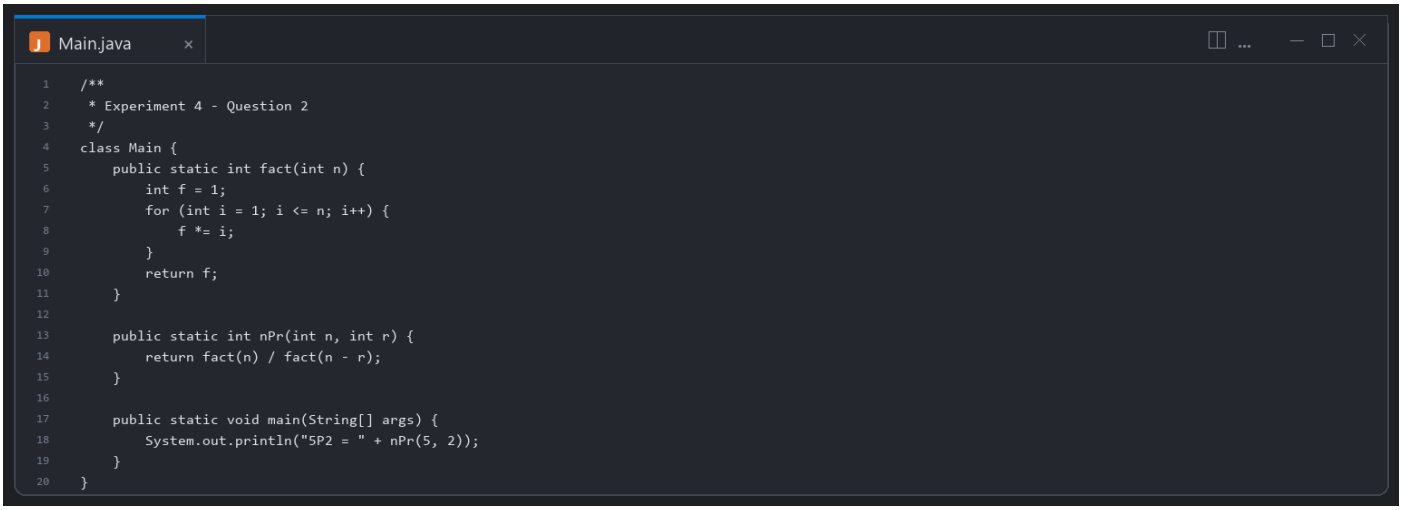


Assignment 4: Writing Functions

Aim: Write functions with parameters and return values, and call them from main.

Question 2: Question2

Code screenshot (IDE-style source view)



Main.java

```java
/**
 * Experiment 4 - Question 2
 */
class Main {
    public static int fact(int n) {
        int f = 1;
        for (int i = 1; i <= n; i++) {
            f *= i;
        }
        return f;
    }

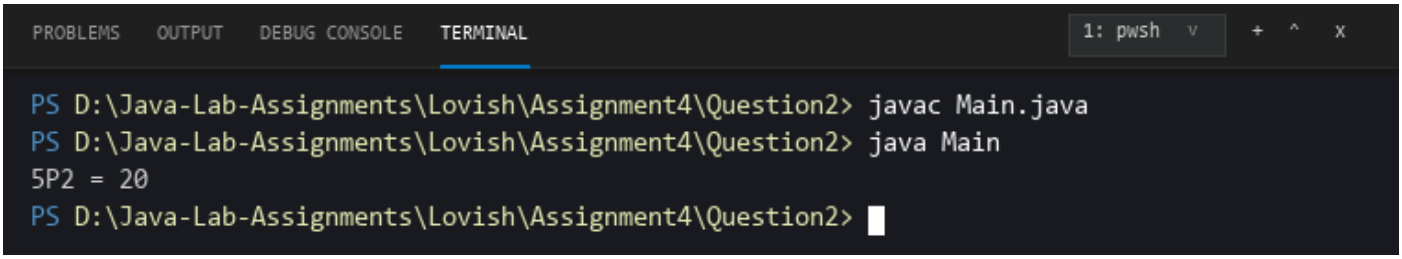
    public static int nPr(int n, int r) {
        return fact(n) / fact(n - r);
    }

    public static void main(String[] args) {
        System.out.println("5P2 = " + nPr(5, 2));
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot



PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

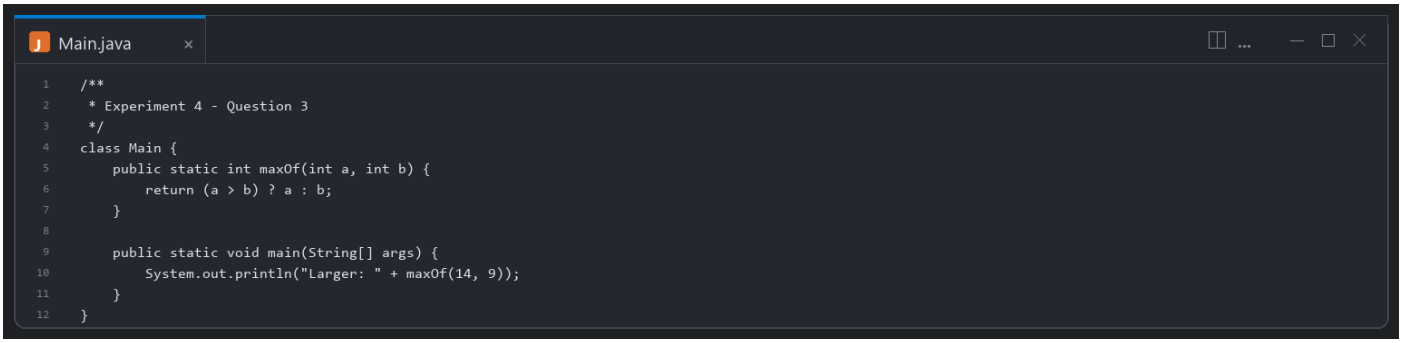
```
PS D:\Java-Lab-Assignments\Lovish\Assignment4\Question2> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment4\Question2> java Main
5P2 = 20
PS D:\Java-Lab-Assignments\Lovish\Assignment4\Question2>
```

Assignment 4: Writing Functions

Aim: Write functions with parameters and return values, and call them from main.

Question 3: Question3

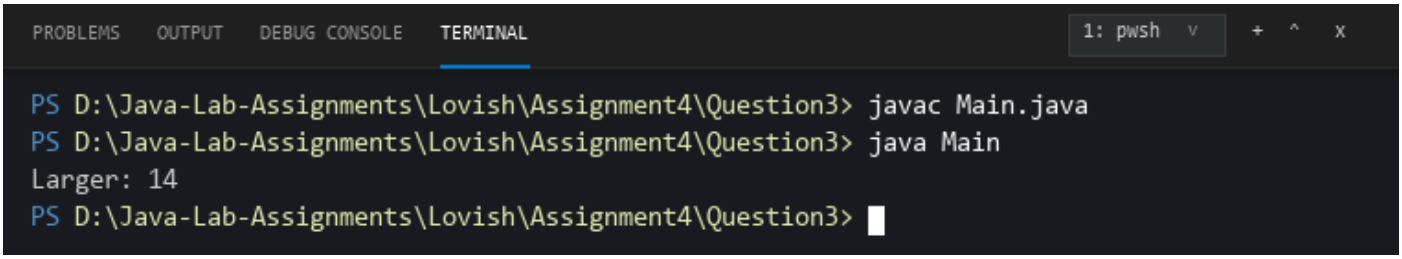
Code screenshot (IDE-style source view)



Input

No runtime input required; the values are initialized in the program.

Output screenshot

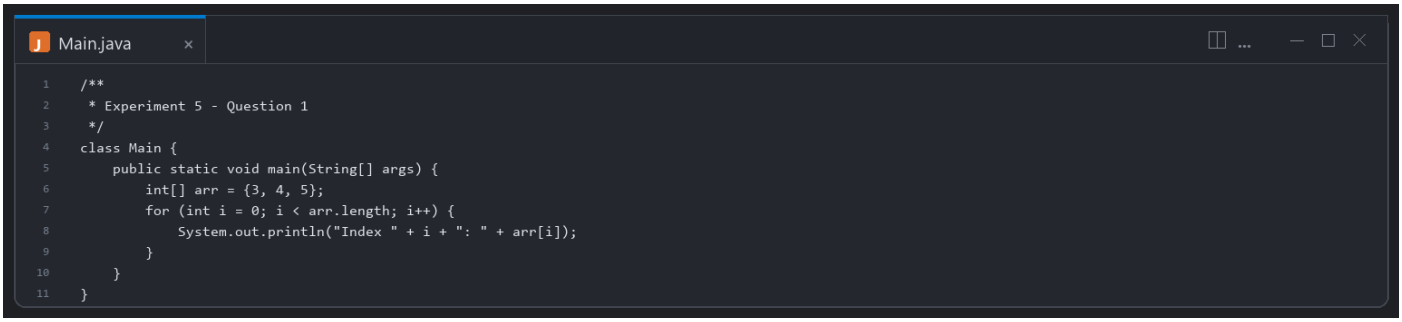


Assignment 5: Arrays: Traversal and Counting

Aim: Create an array, visit every element with a loop, and add up or count the values.

Question 1: Question1

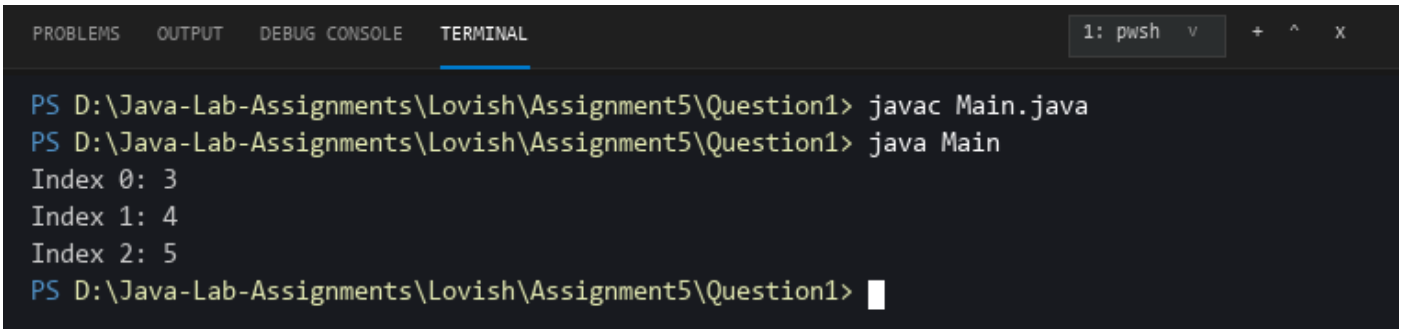
Code screenshot (IDE-style source view)



Input

No runtime input required; the values are initialized in the program.

Output screenshot



Assignment 5: Arrays: Traversal and Counting

Aim: Create an array, visit every element with a loop, and add up or count the values.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 5 - Question 2
 */
class Main {
    public static void main(String[] args) {
        int[] arr = {12, 45, 7, 33, 21};
        int sum = 0;
        int largest = arr[0];
        for (int i = 0; i < arr.length; i++) {
            sum += arr[i];
            if (arr[i] > largest) {
                largest = arr[i];
            }
        }
        System.out.println("Sum: " + sum);
        System.out.println("Largest: " + largest);
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL        1: pwsh

PS D:\Java-Lab-Assignments\Lovish\Assignment5\Question2> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment5\Question2> java Main
Sum: 118
Largest: 45
PS D:\Java-Lab-Assignments\Lovish\Assignment5\Question2>
```

Assignment 5: Arrays: Traversal and Counting

Aim: Create an array, visit every element with a loop, and add up or count the values.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 5 - Question 3
 */
class Main {
    public static void main(String[] args) {
        int[] arr = {5, 12, 2, 14, 7};
        int count = 0;
        for (int i = 0; i < arr.length; i++) {
            if (arr[i] > 10) {
                count++;
            }
        }
        System.out.println("Count greater than 10: " + count);
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

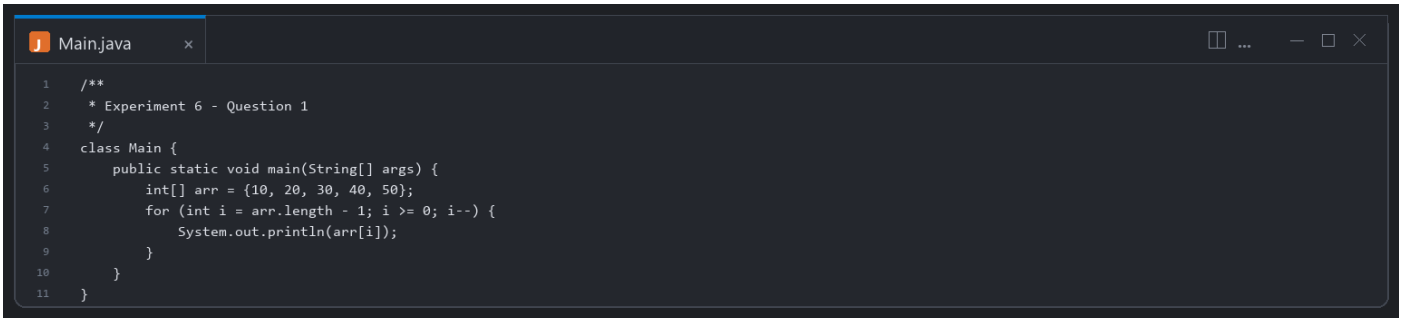
```
PS D:\Java-Lab-Assignments\Lovish\Assignment5\Question3> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment5\Question3> java Main
Count greater than 10: 2
PS D:\Java-Lab-Assignments\Lovish\Assignment5\Question3>
```

Assignment 6: Arrays: Reverse and Duplicates

Aim: Print an array in reverse order, find the smallest value, and check an array for repeated values.

Question 1: Question1

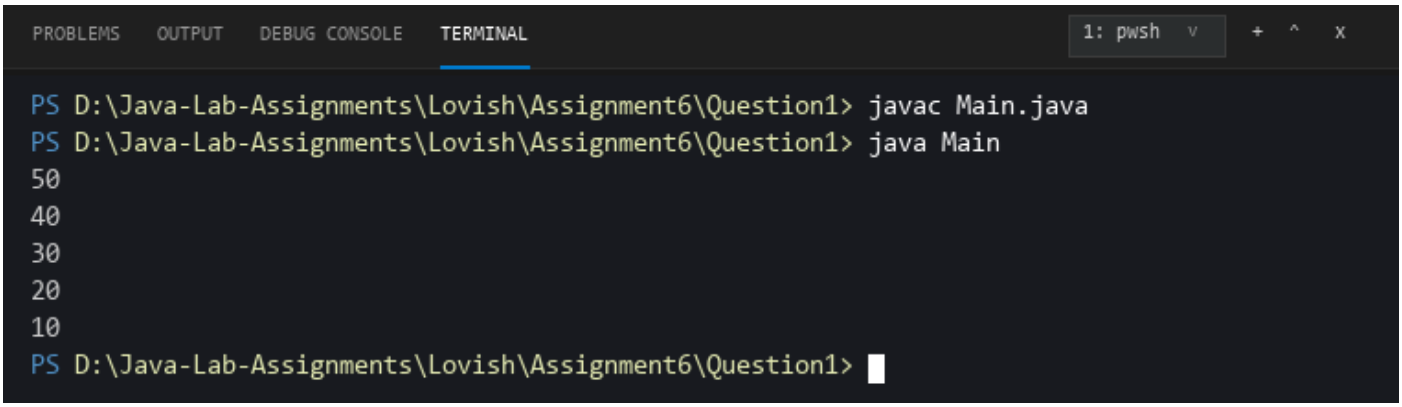
Code screenshot (IDE-style source view)



Input

No runtime input required; the values are initialized in the program.

Output screenshot



Assignment 6: Arrays: Reverse and Duplicates

Aim: Print an array in reverse order, find the smallest value, and check an array for repeated values.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 6 - Question 2
 */
class Main {
    public static void main(String[] args) {
        int[] arr = {2, 5, 1, 8, 3, 4};
        int smallest = arr[0];
        int index = 0;
        for (int i = 1; i < arr.length; i++) {
            if (arr[i] < smallest) {
                smallest = arr[i];
                index = i;
            }
        }
        System.out.println("Smallest: " + smallest + " at index " + index);
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment6\Question2> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment6\Question2> java Main
Smallest: 1 at index 2
PS D:\Java-Lab-Assignments\Lovish\Assignment6\Question2>
```

Assignment 6: Arrays: Reverse and Duplicates

Aim: Print an array in reverse order, find the smallest value, and check an array for repeated values.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 6 - Question 3
 */
class Main {
    public static void main(String[] args) {
        int[] arr = {4, 9, 2, 9, 7};
        boolean found = false;
        int repeated = 0;
        for (int i = 0; i < arr.length; i++) {
            for (int j = i + 1; j < arr.length; j++) {
                if (arr[i] == arr[j]) {
                    repeated = arr[i];
                    found = true;
                    break;
                }
            }
            if (found) break;
        }
        if (found) {
            System.out.println("Repeated value: " + repeated);
        } else {
            System.out.println("No repeated value");
        }
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL                    1: pwsh  +  ^  X

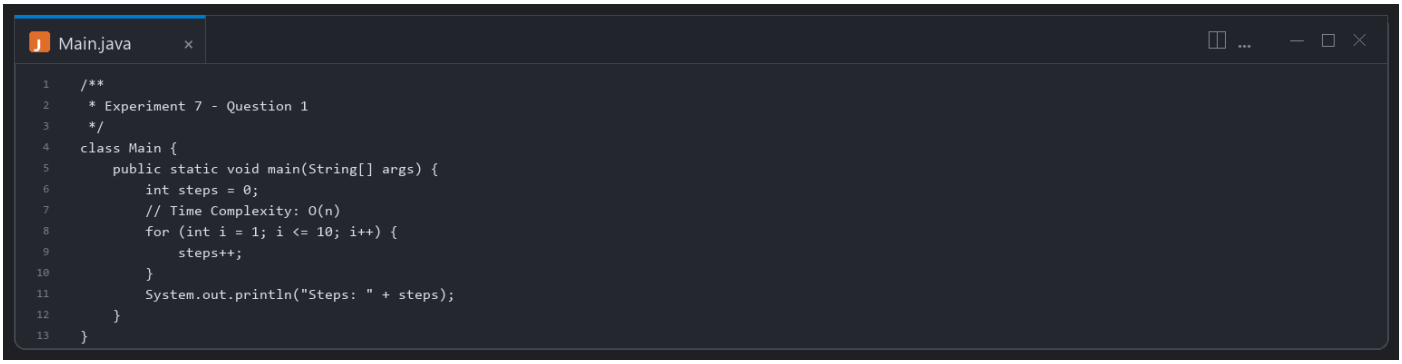
PS D:\Java-Lab-Assignments\Lovish\Assignment6\Question3> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment6\Question3> java Main
Repeated value: 9
PS D:\Java-Lab-Assignments\Lovish\Assignment6\Question3>
```

Assignment 7: Counting Steps and Complexity

Aim: Count the steps a program takes, and write its time complexity as a comment.

Question 1: Question1

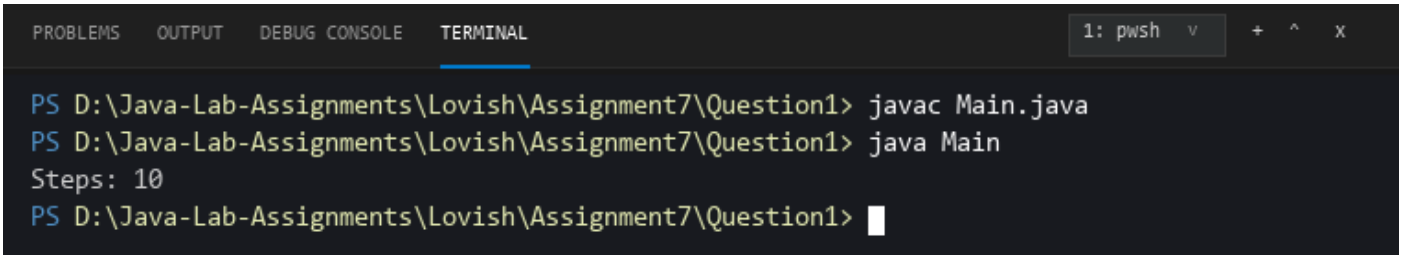
Code screenshot (IDE-style source view)



Input

No runtime input required; the values are initialized in the program.

Output screenshot



Assignment 7: Counting Steps and Complexity

Aim: Count the steps a program takes, and write its time complexity as a comment.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 7 - Question 2
 */
class Main {
    public static void main(String[] args) {
        int steps = 0;
        // Time Complexity: O(n^2)
        for (int i = 1; i <= 10; i++) {
            for (int j = 1; j <= 10; j++) {
                steps++;
            }
        }
        System.out.println("Steps: " + steps);
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment7\Question2> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment7\Question2> java Main
Steps: 100
PS D:\Java-Lab-Assignments\Lovish\Assignment7\Question2>
```

Assignment 7: Counting Steps and Complexity

Aim: Count the steps a program takes, and write its time complexity as a comment.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 7 - Question 3
 */
class Main {
    public static void main(String[] args) {
        int n = 1024;
        int count = 0;
        // Time Complexity: O(log n)
        while (n > 1) {
            n = n / 2;
            count++;
        }
        System.out.println("Halving steps: " + count);
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment7\Question3> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment7\Question3> java Main
Halving steps: 10
PS D:\Java-Lab-Assignments\Lovish\Assignment7\Question3>
```

Assignment 8: Matrices: Transpose and Row Sums

Aim: Create two-dimensional arrays, visit them row by row, and build a new matrix from one.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 8 - Question 1
 */
class Main {
    public static void main(String[] args) {
        int[][] m = new int[3][4];
        int val = 1;
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 4; j++) {
                m[i][j] = val++;
            }
        }
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 4; j++) {
                System.out.print(m[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment8\Question1> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment8\Question1> java Main
1 2 3 4
5 6 7 8
9 10 11 12
PS D:\Java-Lab-Assignments\Lovish\Assignment8\Question1>
```

Assignment 8: Matrices: Transpose and Row Sums

Aim: Create two-dimensional arrays, visit them row by row, and build a new matrix from one.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 8 - Question 2
 */
class Main {
    public static void main(String[] args) {
        int[][] matrix = {
            {1, 2, 3},
            {4, 5, 6},
            {7, 8, 9}
        };
        int[][] transpose = new int[3][3];
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                transpose[j][i] = matrix[i][j];
            }
        }
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(transpose[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment8\Question2> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment8\Question2> java Main
1 4 7
2 5 8
3 6 9
PS D:\Java-Lab-Assignments\Lovish\Assignment8\Question2>
```

Assignment 8: Matrices: Transpose and Row Sums

Aim: Create two-dimensional arrays, visit them row by row, and build a new matrix from one.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 8 - Question 3
 */
class Main {
    public static void main(String[] args) {
        int[][] matrix = {
            {1, 2, 3},
            {4, 5, 6},
            {7, 8, 9}
        };
        for (int i = 0; i < 3; i++) {
            int sum = 0;
            for (int j = 0; j < 3; j++) {
                sum += matrix[i][j];
            }
            System.out.println("Row " + (i + 1) + " sum: " + sum);
        }
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment8\Question3> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment8\Question3> java Main
Row 1 sum: 6
Row 2 sum: 15
Row 3 sum: 24
PS D:\Java-Lab-Assignments\Lovish\Assignment8\Question3>
```

Assignment 9: Matrices: Rows and Columns

Aim: Visit the rows and columns of a matrix with nested loops to print, add and count its values.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 9 - Question 1
 */
class Main {
    public static void main(String[] args) {
        int[][] matrix = {
            {1, 2, 3},
            {4, 5, 6},
            {7, 8, 9}
        };
        System.out.print("First row:");
        for (int j = 0; j < 3; j++) {
            System.out.print(" " + matrix[0][j]);
        }
        System.out.println();
        System.out.print("Last row:");
        for (int j = 0; j < 3; j++) {
            System.out.print(" " + matrix[2][j]);
        }
        System.out.println();
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment9\Question1> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment9\Question1> java Main
First row: 1 2 3
Last row: 7 8 9
PS D:\Java-Lab-Assignments\Lovish\Assignment9\Question1>
```

Assignment 9: Matrices: Rows and Columns

Aim: Visit the rows and columns of a matrix with nested loops to print, add and count its values.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 9 - Question 2
 */
class Main {
    public static void main(String[] args) {
        int[][] matrix = {
            {1, 2, 3},
            {4, 5, 6},
            {7, 8, 9}
        };
        for (int j = 0; j < 3; j++) {
            int sum = 0;
            for (int i = 0; i < 3; i++) {
                sum += matrix[i][j];
            }
            System.out.println("Column " + (j + 1) + " sum: " + sum);
        }
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment9\Question2> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment9\Question2> java Main
Column 1 sum: 12
Column 2 sum: 15
Column 3 sum: 18
PS D:\Java-Lab-Assignments\Lovish\Assignment9\Question2>
```

Assignment 9: Matrices: Rows and Columns

Aim: Visit the rows and columns of a matrix with nested loops to print, add and count its values.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 9 - Question 3
 */
class Main {
    public static void main(String[] args) {
        int[][] matrix = {
            {1, 2, 3},
            {4, 5, 6},
            {7, 8, 9}
        };
        int count = 0;
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                if (matrix[i][j] > 5) {
                    count++;
                }
            }
        }
        System.out.println("Values greater than 5: " + count);
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL        1: pwsh

PS D:\Java-Lab-Assignments\Lovish\Assignment9\Question3> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment9\Question3> java Main
Values greater than 5: 4
PS D:\Java-Lab-Assignments\Lovish\Assignment9\Question3>
```

Assignment 10: Switch and ArrayList

Aim: Choose a branch with a switch statement, and store and change a list of numbers in an ArrayList.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 10 - Question 1
 */
class Main {
    public static void main(String[] args) {
        int day = 3;
        switch (day) {
            case 1:
                System.out.println("Monday");
                break;
            case 2:
                System.out.println("Tuesday");
                break;
            case 3:
                System.out.println("Wednesday");
                break;
            default:
                System.out.println("Invalid day");
                break;
        }
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment10\Question1> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment10\Question1> java Main
Wednesday
PS D:\Java-Lab-Assignments\Lovish\Assignment10\Question1>
```

Assignment 10: Switch and ArrayList

Aim: Choose a branch with a switch statement, and store and change a list of numbers in an ArrayList.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 10 - Question 2
 */
import java.util.ArrayList;

class Main {
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<>();
        list.add(10);
        list.add(20);
        list.add(30);
        list.add(40);

        System.out.println(list);
        System.out.println("Size: " + list.size());
        System.out.println("Item at index 2: " + list.get(2));
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL                1: pwsh

PS D:\Java-Lab-Assignments\Lovish\Assignment10\Question2> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment10\Question2> java Main
[10, 20, 30, 40]
Size: 4
Item at index 2: 30
PS D:\Java-Lab-Assignments\Lovish\Assignment10\Question2>
```

Assignment 10: Switch and ArrayList

Aim: Choose a branch with a switch statement, and store and change a list of numbers in an ArrayList.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 10 - Question 3
 */
import java.util.ArrayList;

class Main {
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<>();
        list.add(10);
        list.add(20);
        list.add(30);
        list.add(40);

        list.set(1, 25);
        list.remove(3);

        System.out.println(list);
        System.out.println("Size: " + list.size());
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL                    1: pwsh  v  +  ^  x
PS D:\Java-Lab-Assignments\Lovish\Assignment10\Question3> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment10\Question3> java Main
[10, 25, 30]
Size: 3
PS D:\Java-Lab-Assignments\Lovish\Assignment10\Question3>
```

Assignment 11: Bubble Sort and Selection Sort

Aim: Sort small arrays with bubble sort and selection sort, and count the swaps.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 11 - Question 1
 */
import java.util.Arrays;

class Main {
    public static void main(String[] args) {
        int[] arr = {5, 1, 4, 2, 8};
        int n = arr.length;
        for (int i = 0; i < n - 1; i++) {
            for (int j = 0; j < n - 1 - i; j++) {
                if (arr[j] > arr[j + 1]) {
                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                }
            }
        }
        System.out.println("Sorted: " + Arrays.toString(arr));
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment11\Question1> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment11\Question1> java Main
Sorted: [1, 2, 4, 5, 8]
PS D:\Java-Lab-Assignments\Lovish\Assignment11\Question1>
```

Assignment 11: Bubble Sort and Selection Sort

Aim: Sort small arrays with bubble sort and selection sort, and count the swaps.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 11 - Question 2
 */
import java.util.Arrays;

class Main {
    public static void main(String[] args) {
        int[] arr = {64, 25, 12, 22, 11};
        int n = arr.length;
        for (int i = 0; i < n - 1; i++) {
            int minIdx = i;
            for (int j = i + 1; j < n; j++) {
                if (arr[j] < arr[minIdx]) {
                    minIdx = j;
                }
            }
            int temp = arr[minIdx];
            arr[minIdx] = arr[i];
            arr[i] = temp;
        }
        System.out.println("Sorted: " + Arrays.toString(arr));
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL                    1: pwsh  v  +  ^  x

PS D:\Java-Lab-Assignments\Lovish\Assignment11\Question2> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment11\Question2> java Main
Sorted: [11, 12, 22, 25, 64]
PS D:\Java-Lab-Assignments\Lovish\Assignment11\Question2>
```

Assignment 11: Bubble Sort and Selection Sort

Aim: Sort small arrays with bubble sort and selection sort, and count the swaps.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 11 - Question 3
 */
class Main {
    public static void main(String[] args) {
        int[] arr = {5, 1, 4, 2, 8};
        int n = arr.length;
        int swaps = 0;
        for (int i = 0; i < n - 1; i++) {
            for (int j = 0; j < n - 1 - i; j++) {
                if (arr[j] > arr[j + 1]) {
                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                    swaps++;
                }
            }
        }
        System.out.println("Swaps: " + swaps);
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL        1: pwsh

PS D:\Java-Lab-Assignments\Lovish\Assignment11\Question3> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment11\Question3> java Main
Swaps: 4
PS D:\Java-Lab-Assignments\Lovish\Assignment11\Question3>
```

Assignment 12: Insertion Sort

Aim: Sort small arrays with insertion sort and count how many values are shifted.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 12 - Question 1
 */
import java.util.Arrays;

class Main {
    public static void main(String[] args) {
        int[] arr = {7, 3, 5, 1};
        for (int i = 1; i < arr.length; i++) {
            int key = arr[i];
            int j = i - 1;
            while (j >= 0 && arr[j] > key) {
                arr[j + 1] = arr[j];
                j--;
            }
            arr[j + 1] = key;
        }
        System.out.println("Sorted: " + Arrays.toString(arr));
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment12\Question1> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment12\Question1> java Main
Sorted: [1, 3, 5, 7]
PS D:\Java-Lab-Assignments\Lovish\Assignment12\Question1>
```

Assignment 12: Insertion Sort

Aim: Sort small arrays with insertion sort and count how many values are shifted.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 12 - Question 2
 */
import java.util.Arrays;

class Main {
    public static void main(String[] args) {
        int[] arr = {14, 3, 9, 21, 6};
        for (int i = 1; i < arr.length; i++) {
            int key = arr[i];
            int j = i - 1;
            while (j >= 0 && arr[j] > key) {
                arr[j + 1] = arr[j];
                j--;
            }
            arr[j + 1] = key;
        }
        System.out.println("Sorted: " + Arrays.toString(arr));
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment12\Question2> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment12\Question2> java Main
Sorted: [3, 6, 9, 14, 21]
PS D:\Java-Lab-Assignments\Lovish\Assignment12\Question2>
```

Assignment 12: Insertion Sort

Aim: Sort small arrays with insertion sort and count how many values are shifted.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 12 - Question 3
 */
class Main {
    public static void main(String[] args) {
        int[] arr = {6, 5, 4, 3, 2, 1};
        int shifts = 0;
        for (int i = 1; i < arr.length; i++) {
            int key = arr[i];
            int j = i - 1;
            while (j >= 0 && arr[j] > key) {
                arr[j + 1] = arr[j];
                shifts++;
                j--;
            }
            arr[j + 1] = key;
        }
        System.out.println("Shifts: " + shifts);
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL        1: pwsh

PS D:\Java-Lab-Assignments\Lovish\Assignment12\Question3> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment12\Question3> java Main
Shifts: 15
PS D:\Java-Lab-Assignments\Lovish\Assignment12\Question3>
```

Assignment 13: Strings: Methods and Palindromes

Aim: Use common String methods, reverse a word, check a palindrome and print substrings.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 13 - Question 1
 */
class Main {
    public static void main(String[] args) {
        String str = "Programming";
        System.out.println("Length: " + str.length());
        System.out.println("Position of g: " + str.indexOf('g'));
        System.out.println("Upper case: " + str.toUpperCase());
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment13\Question1> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment13\Question1> java Main
Length: 11
Position of g: 3
Upper case: PROGRAMMING
PS D:\Java-Lab-Assignments\Lovish\Assignment13\Question1>
```

Assignment 13: Strings: Methods and Palindromes

Aim: Use common String methods, reverse a word, check a palindrome and print substrings.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 13 - Question 2
 */
class Main {
    public static void main(String[] args) {
        String s = "Java";
        String reversed = "";
        for (int i = s.length() - 1; i >= 0; i--) {
            reversed = reversed + s.charAt(i);
        }
        System.out.println("Reversed: " + reversed);
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL                1: pwsh  v  +  ^  x

PS D:\Java-Lab-Assignments\Lovish\Assignment13\Question2> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment13\Question2> java Main
Reversed: avaJ
PS D:\Java-Lab-Assignments\Lovish\Assignment13\Question2>
```

Assignment 13: Strings: Methods and Palindromes

Aim: Use common String methods, reverse a word, check a palindrome and print substrings.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 13 - Question 3
 */
class Main {
    public static void main(String[] args) {
        String word = "racecar";
        boolean isPalindrome = true;
        int n = word.length();
        for (int i = 0; i < n / 2; i++) {
            if (word.charAt(i) != word.charAt(n - 1 - i)) {
                isPalindrome = false;
                break;
            }
        }
        if (isPalindrome) {
            System.out.println(word + " is a palindrome");
        } else {
            System.out.println(word + " is not a palindrome");
        }
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment13\Question3> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment13\Question3> java Main
racecar is a palindrome
PS D:\Java-Lab-Assignments\Lovish\Assignment13\Question3>
```

Assignment 13: Strings: Methods and Palindromes

Aim: Use common String methods, reverse a word, check a palindrome and print substrings.

Question 4: Question4

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 13 - Question 4
 */
class Main {
    public static void main(String[] args) {
        String s = "abc";
        for (int i = 0; i < s.length(); i++) {
            for (int j = i + 1; j <= s.length(); j++) {
                System.out.println(s.substring(i, j));
            }
        }
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment13\Question4> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment13\Question4> java Main
a
ab
abc
b
bc
c
PS D:\Java-Lab-Assignments\Lovish\Assignment13\Question4>
```

Assignment 14: Characters: Counting and Positions

Aim: Use charAt to count vowels and a letter, and print each letter with its position.

Question 1: Question1

Code screenshot (IDE-style source view)



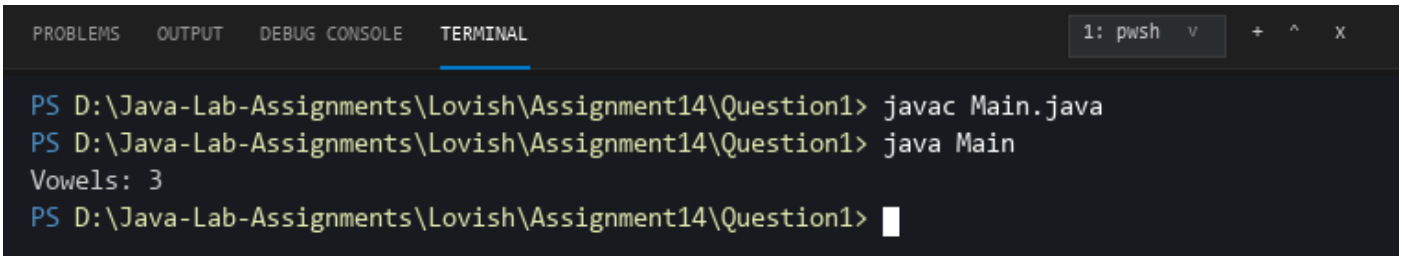
Main.java

```java
/**
 * Experiment 14 - Question 1
 */
class Main {
    public static void main(String[] args) {
        String s = "programming";
        int vowels = 0;
        for (int i = 0; i < s.length(); i++) {
            char ch = Character.toLowerCase(s.charAt(i));
            if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
                vowels++;
            }
        }
        System.out.println("Vowels: " + vowels);
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot



PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment14\Question1> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment14\Question1> java Main
Vowels: 3
PS D:\Java-Lab-Assignments\Lovish\Assignment14\Question1>
```

Assignment 14: Characters: Counting and Positions

Aim: Use charAt to count vowels and a letter, and print each letter with its position.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 14 - Question 2
 */
class Main {
    public static void main(String[] args) {
        String s = "programming";
        int count = 0;
        for (int i = 0; i < s.length(); i++) {
            if (s.charAt(i) == 'g') {
                count++;
            }
        }
        System.out.println("Count of g: " + count);
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment14\Question2> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment14\Question2> java Main
Count of g: 2
PS D:\Java-Lab-Assignments\Lovish\Assignment14\Question2>
```

Assignment 14: Characters: Counting and Positions

Aim: Use charAt to count vowels and a letter, and print each letter with its position.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 14 - Question 3
 */
class Main {
    public static void main(String[] args) {
        String s = "hello";
        for (int i = 0; i < s.length(); i++) {
            System.out.println(i + ": " + s.charAt(i));
        }
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment14\Question3> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment14\Question3> java Main
0: h
1: e
2: l
3: l
4: o
PS D:\Java-Lab-Assignments\Lovish\Assignment14\Question3>
```

Assignment 15: StringBuilder Methods

Aim: Build and change text with StringBuilder, using append, insert and deleteCharAt.

Question 1: Question1

Code screenshot (IDE-style source view)

Main.java

```
/**
 * Experiment 15 - Question 1
 */
class Main {
    public static void main(String[] args) {
        StringBuilder sb = new StringBuilder("Hello");
        sb.append(" World");
        System.out.println(sb);
        System.out.println("Length: " + sb.length());
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment15\Question1> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment15\Question1> java Main
Hello World
Length: 11
PS D:\Java-Lab-Assignments\Lovish\Assignment15\Question1>
```

Assignment 15: StringBuilder Methods

Aim: Build and change text with StringBuilder, using append, insert and deleteCharAt.

Question 2: Question2

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 15 - Question 2
 */
class Main {
    public static void main(String[] args) {
        StringBuilder sb = new StringBuilder();
        for (int i = 1; i <= 5; i++) {
            sb.append(i).append(",");
        }
        sb.deleteCharAt(sb.length() - 1);
        System.out.println(sb);
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment15\Question2> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment15\Question2> java Main
1,2,3,4,5
PS D:\Java-Lab-Assignments\Lovish\Assignment15\Question2>
```

Assignment 15: StringBuilder Methods

Aim: Build and change text with StringBuilder, using append, insert and deleteCharAt.

Question 3: Question3

Code screenshot (IDE-style source view)

Main.java

```java
/**
 * Experiment 15 - Question 3
 */
class Main {
    public static void main(String[] args) {
        StringBuilder sb = new StringBuilder("Java");
        sb.insert(0, "Hello ");
        System.out.println(sb);
    }
}
```

Input

No runtime input required; the values are initialized in the program.

Output screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 1: pwsh

```
PS D:\Java-Lab-Assignments\Lovish\Assignment15\Question3> javac Main.java
PS D:\Java-Lab-Assignments\Lovish\Assignment15\Question3> java Main
Hello Java
PS D:\Java-Lab-Assignments\Lovish\Assignment15\Question3>
```